A Scoped Zero-Thickness S4 Tetrahedral Hinge-Tree Certificate: Median-Plane Geometry, One-Parameter Route Wrappers, and a Selected-Hinge Contact Bridge

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Abstract

We give a scoped, reproducible certificate package for a specific four-piece ("S4") dissection of the regular tetrahedron by two opposite-edge median planes. The static dissection consists of four congruent tetrahedra whose closed union is the ambient tetrahedron. The moving model is a zero-thickness rigid-piece hinge-tree model on two audited representative signed rays, `TREE_007` and `TREE_021`, with parameter domain

$$\theta = 0 \quad \text{and} \quad 0 < \theta \leq 120^\circ.$$

At $\theta = 0$, the pieces form a closed-contact tiling with no strict interior overlap. On the open ray, every unordered piece-pair obligation in the two representative trees is routed to one of three certified predicate layers: six selected-hinge B04 rows, four common-edge B05 rows, and two shared-face B06/B07 rows; route-clean B03 obligations are vacuous on this one-parameter spine.

The main theorem is intentionally narrow. It is not a proof of physical hingeability, positive-thickness printability, global collision-free motion, or three-parameter bounded-cell closure. The nontrivial semantic point is the selected-hinge layer: an A7c signed-orientation certificate is not used alone. A package-local B04/A7c bridge records six row-level wedge certificates showing that the selected opening side implies boundary hinge contact and no strict local interior overlap at the contact plane, excluding the hinge axis. The repository also includes RW12, a corrected body-preserving digital fabrication handoff for `TREE_007`; RW12 is supplementary digital material and not physical validation.

Keywords. regular tetrahedron; hinged mechanisms; rigid-piece kinematics; computational certificates; Sturm certificates; route wrappers; digital fabrication handoff.

Companion addendum note. This repository also contains a promoted companion addendum for the equal-magnitude zero-thickness S4 theta-star finite atlas under `extensions/theta_star_finite_atlas/`. The addendum has its own TeX/PDF, bibliography, proof-spine package, and reproducibility gates, and it passed external mathematical red-team review in its stated scope. It is not used in the theorem proved below and does not broaden the public claim boundary of this main paper.

1 Introduction

This paper is a certificate paper for a fixed geometric object, not a general hinged-dissection theorem. The object is the median-plane S4 dissection of a regular tetrahedron. The four pieces are tetrahedra; at the closed endpoint they tile the original tetrahedron, and along two audited one-parameter signed rays they are moved by rooted rigid hinge-tree kinematics.

The certificate is useful precisely because the tempting stronger statements are false or unproved. The package does not prove that a positive-thickness printable mechanism exists. It does not prove that every possible S4 hinge choice works. It does not prove dynamic connectedness between the two audited representative signed rays. It proves a scoped statement for two representative trees and a one-dimensional angular domain, with every pair obligation routed to a recorded certificate layer.

Contribution. The package contributes four pieces of evidence.

1. An exact coordinate model for the S4 median-plane dissection and its closed-contact endpoint.
2. A rooted rigid-hinge kinematic model for `TREE_007` and `TREE_021` with explicit hinge axes and signed-ray conventions.
3. A route-wrapper theorem on the domain $\theta = 0$ plus $0 < \theta \leq 120^\circ$, supported by the A6–A7d certified layers.
4. A local B04/A7c bridge that upgrades six selected-hinge signed rows from mere orientation signs to contact-side/no-strict-local-overlap records.

Replay-backed claims are labelled as *Certified Propositions*. These are finite or symbolic certificate statements tied to files in the repository, not claims of a new general theorem in the literature.

Public boundary. The claim boundary is part of the result. The package allows the zero-thickness one-parameter route-wrapper claim and the six-row selected-hinge bridge claim. It disallows physical validation, positive-thickness hinge validity, global collision-free motion, and three-parameter bounded-cell closure. These boundaries are mirrored in `docs/PUBLIC_CLAIM_BOUNDARY.md`, `docs/CLAIM_LEDGER.md`, and `docs/PROOF_OBLIGATIONS.md`.

Organization. Section 2 gives the exact tetrahedral model. Section 3 records the closed endpoint and kinematic conventions. Section 4 states the scoped one-parameter theorem wrapper. Section 5 explains the A6–A7d certificate layers. Section 6 proves the selected-hinge wedge bridge used by the B04 rows. Sections 7 and 8 describe reproducibility and the RW12 digital-fabrication boundary. Section 9 places the result next to related hinged mechanism and fabrication work.

2 The S4 median-plane model

Let $\mathcal{T} = \operatorname{conv}(A, B, C, D) \subset \mathbb{R}^3$ be the unit-edge regular tetrahedron with

$$A = (0,0,0), \quad B = (1,0,0), \quad C = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}, 0\right), \quad D = \left(\frac{1}{2}, \frac{\sqrt{3}}{6}, \frac{\sqrt{6}}{3}\right).$$

Let

$$M_{AB} = \frac{A+B}{2} = \left(\frac{1}{2}, 0, 0\right), \qquad M_{CD} = \frac{C+D}{2} = \left(\frac{1}{2}, \frac{\sqrt{3}}{3}, \frac{\sqrt{6}}{6}\right).$$

The S4 pieces are the four closed tetrahedra

$$P_0 = \operatorname{conv}(A, M_{AB}, C, M_{CD}), \quad P_1 = \operatorname{conv}(A, M_{AB}, D, M_{CD}),$$
$$P_2 = \operatorname{conv}(B, M_{AB}, C, M_{CD}), \quad P_3 = \operatorname{conv}(B, M_{AB}, D, M_{CD}).$$

Lemma 2.1 (Median-plane S4 tiling)**.** *The tetrahedron $\mathcal{T}$ is partitioned by the two barycentric median planes $\alpha = \beta$ and $\gamma = \delta$ into the four closed pieces P_0, P_1, P_2, P_3. Their relative interiors are pairwise disjoint, their closed union is $\mathcal{T}$, and each piece has volume $\sqrt{2}/48$. The four pieces are congruent tetrahedra with edge-length multiset*

$$\left\{\frac{1}{2}, \frac{1}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{2}, 1\right\}.$$

Proof. Write a point of $\mathcal{T}$ in barycentric coordinates

$$x = \alpha A + \beta B + \gamma C + \delta D, \qquad \alpha, \beta, \gamma, \delta \geq 0, \qquad \alpha + \beta + \gamma + \delta = 1.$$

The plane through C, D, M_{AB} is $\alpha = \beta$, and the plane through A, B, M_{CD} is $\gamma = \delta$. For example, on the cell $\alpha \geq \beta$, $\gamma \geq \delta$, one has

$$x = (\alpha - \beta)A + 2\beta M_{AB} + (\gamma - \delta)C + 2\delta M_{CD},$$

with nonnegative coefficients summing to one, hence $x \in P_0$. The other three sign cells give the analogous formulas for P_1, P_2, P_3. Opposite strict sign inequalities cannot hold simultaneously, so relative interiors are disjoint.

The volume calculation

$$\mathrm{Vol}(P_0) = \frac{1}{6} \left|\det(M_{AB} - A, C - A, M_{CD} - A)\right| = \frac{\sqrt{2}}{48}$$

applies the same volume for the other pieces by the ambient symmetries interchanging A, B and C, D. The listed edge lengths follow by direct coordinate differences for P_0, and symmetry gives the same multiset for all pieces. □

The closed pair-contact ledger is fixed throughout the package.

Pair	Contact ID	Contact type	Shared set
$P_0 - P_1$	C0	shared face	$\mathrm{conv}(A, M_{AB}, M_{CD})$
$P_0 - P_2$	C1	shared face	$\mathrm{conv}(C, M_{AB}, M_{CD})$
$P_0 - P_3$	C2	shared edge	$\mathrm{conv}(M_{AB}, M_{CD})$
$P_1 - P_2$	C3	shared edge	$\mathrm{conv}(M_{AB}, M_{CD})$
$P_1 - P_3$	C4	shared face	$\mathrm{conv}(D, M_{AB}, M_{CD})$
$P_2 - P_3$	C5	shared face	$\mathrm{conv}(B, M_{AB}, M_{CD})$

Certified Proposition 2.2 (Source-lock for notation)**.** *The coordinate, piece, pair, contact, and hinge labels in this paper are locked by* `certified/source_docs/S4_LEMMA_00_DEFINITIONS_AND_NOTATION_LOCK.md` *and* `certified/source_docs/S4_LEMMA_01_GEOMETRY_AND_TILING.md`*. Later certificate rows are interpreted using these labels.*

3 Closed endpoint and rooted hinge-tree kinematics

The endpoint $\theta = 0$ is a closed-contact endpoint, not a positive-clearance endpoint. The pieces are closed sets and are allowed to meet along the catalogued contacts in Section 2.

Lemma 3.1 (Closed-contact endpoint)**.** *At $\theta = 0$, the four pieces lie in $\mathcal{T}$, their volumes sum to $\mathrm{Vol}(\mathcal{T})$, all six unordered pairs are exactly the catalogued contacts C0–C5, and no two distinct pieces have strict interior overlap. This endpoint has zero clearance at the catalogued contacts.*

Proof. By Lemma 2.1, the closed union is $\mathcal{T}$ and relative interiors are pairwise disjoint. Thus every pairwise intersection is contained in a proper face or edge of each piece and cannot contain a three-dimensional interior point common to two pieces. The contact table in Section 2 lists all six unordered piece pairs. Since the list includes shared faces and shared edges, the endpoint is contact-allowing and not positive-clearance. □

The moving model uses rooted hinge trees. For an oriented axis from a to b, write $u = (b - a)/\|b - a\|$. The rotation about the affine line through a, b by angle ϕ is

$$\mathrm{Rot}(a, b, \phi)(x) = Q_u(\phi)x + a - Q_u(\phi)a,$$

where Q_u is Rodrigues' rotation matrix. Points on the axis are fixed.

The audited representative trees are:

Tree	Hinge graph	Signed hinge angles
TREE_007	P_0P_1, P_0P_2, P_1P_3	$d_{H0} = \theta,\ d_{H4} = \theta,\ d_{H7} = -\theta$
TREE_021	P_0P_1, P_1P_3, P_2P_3	$d_{H0} = \theta,\ d_{H7} = -\theta,\ d_{H9} = \theta$

The root is P_0 in both cases.

Lemma 3.2 (Well-defined rooted kinematics)**.** *For TREE_007 and TREE_021, any real assignment of signed angles to the selected hinges determines a unique rigid transform F_i for each piece P_i. For each selected hinge edge, the parent and child transforms agree on the hinge axis.*

Proof. Each selected graph has four vertices, three edges, and the rooted paths listed in the source lock; hence it is a tree. There is a unique path from root P_0 to every piece. Starting with $F_0 = \mathrm{id}$, propagate along each edge by composing the parent transform with the axis rotation about the parent- placed hinge line. A composition of rigid transforms is rigid, and uniqueness of the rooted path gives uniqueness of F_i. Since $\mathrm{Rot}(a, b, \phi)$ fixes the axis line pointwise, parent and child transforms agree on the selected hinge axis. □

At $\theta = 0$, every signed angle is zero, every axis rotation is the identity, and all piece transforms are identity. Thus both representative trees have the same closed-contact endpoint from Lemma 3.1.

Certified Proposition 3.3 (Kinematic source-lock)**.** *The endpoint and kinematic statements are source-locked to certified/source_docs/S4_LEMMA_02_CLOSED_ENDPOINT.md and certified/source_docs/S4_LEMMA_03_KINEMATICS_AND_SIGNS.md. These files also record the exact hinge IDs H0_A_M_AB, H4_C_M_CD, H7_D_M_CD, and H9_B_M_AB.*

4 The scoped one-parameter wrapper

The one-parameter open domain is

$$0 < \theta \leq 120^\circ.$$

The certificate uses the half-angle coordinate

$$t = \tan(\theta/2), \qquad 0 < t \leq \sqrt{3},$$

and many sign proofs are carried out on the rational open superset

$$0 < t < 2.$$

Every representative tree has six unordered pair obligations. The A7d wrapper routes the twelve obligations across the two trees as follows.

Tree	Pair	Route
TREE_007	$P_0 - P_1$	B04 selected-hinge contact-side
TREE_007	$P_0 - P_2$	B04 selected-hinge contact-side
TREE_007	$P_0 - P_3$	B05 common-edge projection
TREE_007	$P_1 - P_2$	B05 common-edge projection
TREE_007	$P_1 - P_3$	B04 selected-hinge contact-side
TREE_007	$P_2 - P_3$	B06/B07 shared-face residual
TREE_021	$P_0 - P_1$	B04 selected-hinge contact-side
TREE_021	$P_0 - P_2$	B06/B07 shared-face residual
TREE_021	$P_0 - P_3$	B05 common-edge projection
TREE_021	$P_1 - P_2$	B05 common-edge projection
TREE_021	$P_1 - P_3$	B04 selected-hinge contact-side
TREE_021	$P_2 - P_3$	B04 selected-hinge contact-side

Theorem 4.1 (Scoped zero-thickness one-parameter wrapper)**.** *For the catalogued median-plane S4 pieces, the public package records a closed zero-thickness one-parameter wrapper for the two audited representatives `TREE_007` and `TREE_021` on the domain*

$$\theta = 0 \quad \textit{and} \quad 0 < \theta \leq 120^\circ.$$

At $\theta = 0$, the configuration is the closed-contact endpoint of Lemma 3.1. On the open ray, every unordered pair obligation in each representative tree is assigned to a completed certificate layer: six B04 selected-hinge bridge rows, four B05 common-edge projection rows, and two B06/B07 shared-face residual rows. There are no route-clean B03 obligations on this one-parameter spine.

Proof. The endpoint clause follows from Lemmas 2.1–3.2: at $\theta = 0$, all rooted transforms are identity and the pieces are in the catalogued closed tiling.

For the open ray, the A7d manifest is a pair-route ledger. It contains two wrapper records, one for `TREE_007` and one for `TREE_021`, each with six unordered piece-pair obligations. The manifest records the predicate route counts

$$\text{B04} = 6, \qquad \text{B05} = 4, \qquad \text{B06/B07} = 2.$$

The B03 count is zero by the A7b route-vacuity certificate. The routed layers are discharged by the certified propositions in Sections 5 and 6. Thus every open-ray pair obligation is routed to a completed package-local layer, and no additional B03 obligation remains on this scoped one-parameter domain. □

Certified Proposition 4.2 (A7d manifest)**.** *The theorem wrapper above is the statement recorded in `certified/a7d_one_parameter_theorem_wrapper_manifest.json`. The manifest has `record_count=2`, `wrapper_closed_count=2`, and object status count `a7d_one_parameter_ray_wrapper_closed: 2`.*

Remark 4.3 (Scope)*.* The theorem is a route-wrapper theorem for a zero-thickness one-parameter model. It is not a positive-thickness fabrication theorem, not a global collision-free motion theorem, and not a three-parameter bounded-cell theorem.

5 Certificate layers A6–A7d

This section records the certificate layers used by Theorem 4.1. The layers are package-local and source-backed; their role is to discharge the routed pair obligations, not to prove a broader mechanism theorem.

Certified Proposition 5.1 (A6: B05 common-edge projection)**.** *The B05 common-edge projection layer is symbolically closed on the one-parameter ray for the relevant records. A6 packages the A3/A4 and A5 algebraic certificates: A3/A4 proves full-domain raw gap, normalized gap, and axis positivity on the open ray, while A5 splits at $t = 1$ and proves post-switch support signatures and projection gaps on $1 < t < 7/4$, which contains $1 < t \leq \sqrt{3}$. The A7d wrapper uses four B05 pair routes.*

Certification. The source is `certified/source_docs/S4_CL5_A6_ONE_PARAMETER_RAY_CLOSURE_PACKAGE.md`. It records seven symbolic B05 closures in the source package and the one-parameter B05 closure chain used by the wrapper. The public wrapper consumes the B05 role through `certified/a7d_one_parameter_theorem_wrapper_manifest.json`. □

Certified Proposition 5.2 (A7a: B06/B07 shared-face residual)**.** *For the two one-parameter shared-face residual targets,*

$$\textit{TREE\_007}: P_2 - P_3, \qquad \textit{TREE\_021}: P_0 - P_2,$$

the residual shared-face formula is positive on the open ray. In half-angle coordinate the certified rational form is

$$\frac{t^3}{(1+t^2)^2}, \qquad 0 < t < 2.$$

In particular it is positive on $0 < t \leq \sqrt{3}$.

Proof. The trigonometric formula from the source layer is

$$\frac{(1-\cos\theta)\sin\theta}{4} = \sin^3(\theta/2)\cos(\theta/2).$$

With $t = \tan(\theta/2)$, this becomes

$$\frac{t^3}{(1+t^2)^2}.$$

The numerator and denominator have no open roots on $0 < t < 2$; at $t = 1$ both are positive. Therefore the expression is positive on the certified open superset, hence on the audited ray domain. The Sturm/root-count certificate is recorded in `certified/source_docs/S4_CL5_A7A_SHARED_FACE_RESIDUAL_STURM_CERTIFICATE.md`. □

Certified Proposition 5.3 (A7b: B03 route vacuity)**.** *There are no route-clean ordinary non-contact B03 obligations on the audited one-parameter representative rays. Across the two representatives, the twelve pair obligations route as six selected-hinge contacts, four residual shared-edge common-edge projections, and two residual shared-face targets.*

Certification. This is the A7b route-vacuity certificate in `certified/source_docs/S4_CL5_A7B_B03_RAY_VACUITY_CERTIFICATE.md`. It records `one-parameter pair records: 12`, `B03 route-clean pairs: 0`, and `B03 Sturm obligations: 0`. □

Certified Proposition 5.4 (A7c: selected-hinge signed orientation)**.** *For the six selected B04 rows, A7c certifies the sign of the selected hinge angle on the open ray. The signed orientation has the same sign as* $s_h t$ *on* $0 < t < 2$, *with ray-sign counts* $+1 : 4$ *and* $-1 : 2$.

Proof. Lemma 3.2 fixes the signed-ray convention $d_h(\theta) = s_h\theta$. In half-angle coordinates, the relevant signed orientation is sign-equivalent to $s_h t$. The only root is at $t = 0$, which is excluded from the open ray. The sample point $t = 1$ fixes the sign. The six rows are recorded in `certified/a7c_selected_hinge_contact_side_certificate_manifest.json` and `certified/source_docs/S4_CL5_A7C_SELECTED_HINGE_CONTACT_SIDE_CERTIFICATE.md`. □

A7c alone is not the B04 contact-side proof. It is the signed-orientation source layer consumed by the bridge in Section 6.

6 The B04/A7c selected-hinge bridge

The B04 rows are zero-margin selected hinge contacts. They cannot be certified by positive SAT clearance: the pieces are meant to remain joined along the hinge axis. The correct local statement is boundary contact plus no strict local interior overlap away from the hinge axis.

Fix one selected row. Let the oriented hinge axis be $a \to b$, and let F be the third vertex of the source shared face, so the source face is $\mathrm{conv}(a, b, F)$. Let Q_p be the parent apex not in this face and Q_c the child apex not in this face. Define the oriented side form

$$\Omega(X, Y) = (b-a)\cdot((X-a)\times(Y-a)).$$

Lemma 6.1 (Local selected-hinge wedge)**.** *For each of the six selected B04 rows, with A7c ray sign σ, the package records the exact side identities*

$$\sigma\Omega(F, Q_p) = -\frac{\sqrt{2}}{8} < 0, \qquad \sigma\Omega(F, Q_c) = \frac{\sqrt{2}}{8} > 0.$$

Thus the A7c sign identifies the child opening side of the selected source face. Moreover the cross-section angle α between the source face ray and either adjacent tetrahedral apex ray satisfies

$$\cos\alpha = \frac{\sqrt{6}}{3} > \frac{1}{2}, \qquad \textit{so} \qquad 0 < \alpha < 60^\circ.$$

Consequently, for $0 < \theta \leq 120^\circ$, the opened child sector satisfies $\theta + \alpha < 180^\circ$ and cannot wrap behind the hinge axis into the parent sector.

Proof. The identities for Ω are exact coordinate evaluations in the normal cross-section about the selected hinge axis. They place the parent apex and child apex on opposite sides of the oriented source face, after multiplying by the row sign σ. The angle calculation gives $\cos\alpha = \sqrt{6}/3 > 1/2$, hence $\alpha < 60^\circ$. Since the ray domain has $\theta \leq 120^\circ$, the sector opened by the child face remains within an open half-turn from the source face. Therefore it cannot cross through the opposite parent sector away from the hinge axis. □

The bridge also checks the actual rotated child vertices using $t = \tan(\theta/2)$. For the rotated third vertex of the selected source face, the signed side is

$$\frac{t}{2(t^2+1)} > 0, \qquad 0 < t \leq \sqrt{3}.$$

For the rotated child apex, each row reduces to a positive rational form recorded in the row JSON: the denominator is positive, the sample signs at $t = 1$ and $t = \sqrt{3}$ are positive, and all positive numerator roots lie strictly above $\sqrt{3}$.

Certified Proposition 6.2 (B04/A7c bridge rows)**.** *The package-local bridge accepts exactly six records:*

Tree	*Pair*	*Hinge*	*Contact*	*Ray sign*
TREE_007	$P_0 - P_1$	*H0_A_M_AB*	*C0*	$+1$
TREE_007	$P_0 - P_2$	*H4_C_M_CD*	*C1*	$+1$
TREE_007	$P_1 - P_3$	*H7_D_M_CD*	*C4*	-1
TREE_021	$P_0 - P_1$	*H0_A_M_AB*	*C0*	$+1$
TREE_021	$P_1 - P_3$	*H7_D_M_CD*	*C4*	-1
TREE_021	$P_2 - P_3$	*H9_B_M_AB*	*C5*	$+1$

For these rows, A7c signed orientation plus the local wedge bridge proves boundary hinge contact and no strict local interior overlap at the contact plane, excluding the hinge axis.

Certification. The bridge manifest is `certified/b04_a7c_contact_side_bridge_manifest.json`. It records `record_count=6`, `accepted_bridge_record_count=6`, and the policy that bridge rows are consumed from the A7c manifest records while asserting equality of `tree_id`, `hinge_id`, `pair`, and `ray_sign`. The row records live in `certified/a7c_b04_contact_side_bridge_records/`. The derivation and claim boundary are source-locked in `certified/source_docs/S4_CL5_A7C_B04_CONTACT_SIDE_BRIDGE_CERTIFICATE.md`. □

Remark 6.3. The bridge is local and selected-row scoped. It does not give positive separation for selected hinges, does not handle hinge thickness, and does not extend to unselected or three-parameter B04 cases.

7 Public package and reproducibility

The repository is organized as a paper-as-public-package: the manuscript, certificate manifests, row-level records, replay scripts, hash manifest, and claim-boundary ledgers are shipped together. This follows established reproducible-artifact practice rather than asserting a new standard; its role here is to make the computational dependencies auditable while separating the mathematical theorem from supplementary digital-fabrication artifacts [11, 10]. The main paper source and PDF live in `paper/`. Mathematical and certificate controls live in `docs/`, `certified/`, `scripts/`, and `results/`. The public package manifest is `paper/PUBLIC_PACKAGE_MANIFEST.json`.

Certified Proposition 7.1 (Public package gate)**.** *The local public package gate passes. The command*

```
python scripts/run_all_reproducibility_checks.py
```

checks required files, B04/A7c bridge row counts, RW12 hash consistency, manifest hashes, and absence of promoted physical-validation phrases. The current public package check records all checks passing.

Certification. The generated report is `results/public_package_check.json`. The pytest wrapper `tests/test_public_package.py` runs the same package checker. The manifest records `LOCAL_RELEASE_GATES_PASSED`, `public_export_ready=true`, and `physical_validation_claimed=false`. □

The source-lock boundary is intentionally duplicated: the paper cites concise Certified Propositions, while the repository ships the longer source documents under `certified/source_docs/`. In particular, the public package includes S4 Lemmas 00–12 and the B04 contact-orientation review notes, so the bridge source dependencies are present in the package rather than only in the private workspace.

Remark 7.2 (What the tests do not prove). The public tests verify package integrity and row-level consistency. They do not replace the mathematical content of Lemmas 2.1–6.1 and Theorem 4.1; they keep the shipped artifacts synchronized with the claim boundary.

8 RW12 supplementary digital fabrication handoff

RW12 is included because it is practically useful, not because it strengthens Theorem 4.1. It is a corrected body-preserving digital handoff for `TREE_007` containing fabrication-oriented files, checklist material, and a compact package zip. Its governing report is `results/rw12_external_fabrication_review_package/rw12_external_fabrication_review_report.json`.

Certified Proposition 8.1 (RW12 hash and claim boundary)**.** *The public checker verifies that the RW12 zip hash matches the RW12 report and that RW12 does not promote a physical-validation claim. The RW12 package is therefore an appendix-level digital handoff, not theorem evidence.*

The practical interpretation is straightforward: RW12 can be passed to a CAD, fabrication, or external-review workflow as a digital artifact. It does not certify positive-thickness hinge clearances, material tolerances, assembly success, printed prototype behavior, or measured motion.

Remark 8.2. This distinction matters for publication. The mathematical result is the zero-thickness one-parameter certificate package. A real-world mechanism would require additional positive-thickness modelling, tolerances, fabrication measurements, and physical validation. Those are future work, not hidden assumptions of the present theorem.

9 Related work and remaining obligations

The general hinged-dissection landscape is much broader than this paper. Hinged dissections exist in substantial generality [1]; reversible hinged dissections and reversible nets are studied in [2, 3]; and standard background on geometric folding, linkages, and polyhedra is given by Demaine and O'Rourke [7]. Continuous flattening, vertex-unfolding, and digital jointed assembly systems provide adjacent contexts [5, 6, 9, 8].

The concrete object of this paper — the S_4 median-plane quadrisection of the regular tetrahedron — is the $n = 4$ member of the congruent-dissection atlas of [4], which poses explicitly the question of whether a known congruent dissection can be realised as a physical hinged mechanism (naming S_2 and S_4 as natural edge-hinge candidates). The present certificate treats a *zero-thickness relaxation* of the S_4 case that is relevant to that question; it does *not* resolve the physical hingeability question (positive thickness, printable mechanism), which this paper explicitly leaves open.

The novelty candidate here is not a new universal hinged-dissection theorem. It is the combination of a fixed tetrahedral median-plane model, exact zero-thickness rooted hinge-tree kinematics, row-level route-wrapper certificates, and a public companion repository that keeps theorem evidence separate from digital fabrication artifacts.

Remaining obligations. The following are deliberately open or out of scope.

- Positive-thickness hinge design and physical fabrication validation.
- Global collision-free motion beyond the audited one-parameter domain.
- Dynamic connectedness between `TREE_007` and `TREE_021`.
- Three-parameter bounded-cell closure.
- Extension of the B04 bridge beyond the six selected rows.
- A general theorem for all S4 hinge trees or all tetrahedral dissections.

The package is therefore best read as a scoped certificate paper: it identifies what has been proved for a concrete object, ships the evidence in a reproducible layout, and makes the boundaries visible enough for independent review.

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